

Final Project 3 : E-commerce Application

CI/CD Deployment & Monitoring for React Application

This repository contains a containerized React application and a fully automated CI/CD pipeline. The infrastructure is hosted on AWS EC2, with Jenkins orchestrating the build and deployment process, and a Prometheus/Grafana stack handling uptime monitoring and alerting.

Technology Stack

- **Frontend:** React.js
- **Containerization:** Docker & Docker Compose
- **CI/CD:** Jenkins, GitHub Webhooks, Bash Scripting
- **Registry:** Docker Hub (Public dev repo, Private prod repo)
- **Infrastructure:** AWS EC2 (Ubuntu 24.04 LTS)
- **Monitoring & Alerting:** Prometheus, Grafana, Alertmanager, Blackbox Exporter

Architecture & Workflow

The following diagram illustrates the complete CI/CD lifecycle, from code commit to production deployment and monitoring.

```
```mermaid
```

```
graph TD
```

```
%% Actors and VCS
```

```
Dev((Developer)) -->|Push Code| GitHub[GitHub Repository
dev / master]
```

```
%% CI/CD Pipeline
```

```
GitHub -->|Webhook Trigger| Jenkins[Jenkins CI/CD]
```

```
subgraph CI/CD Process
```

```
 Jenkins -->|1. Execute build.sh| Build[Build Docker Image]
```

```
 Jenkins -->|2. Push| DockerHub[(Docker Hub)]
```

```
 Jenkins -->|3. Execute deploy.sh| EC2_Deploy[Deploy to EC2]
```

```
end
```

```
%% Registry
```

```
DockerHub -->|Public Image| Repo_Dev[Dev Repository]
```

```
DockerHub -->|Private Image| Repo_Prod[Prod Repository]
```

```
%% AWS Environment
```

```
subgraph AWS EC2 Instance (Ubuntu 24.04)
```

```
 EC2_Deploy --> App[React App
Port 80]
```

```
%% Monitoring Stack
```

```
 App_Probe[Blackbox Exporter
Port 9115] -->|HTTP 2xx Probe| App
```

```
 Prometheus[Prometheus
Port 9090] -->|Scrape Metrics| App_Probe
```

Grafana[Grafana<br/>Port 3000] -->|Query Data| Prometheus  
Prometheus -->|Fire Alerts| Alertmanager[Alertmanager<br/>Port 9093]

end

%% External Endpoints

Alertmanager -->|Gmail SMTP| Email[Admin Email Notification]

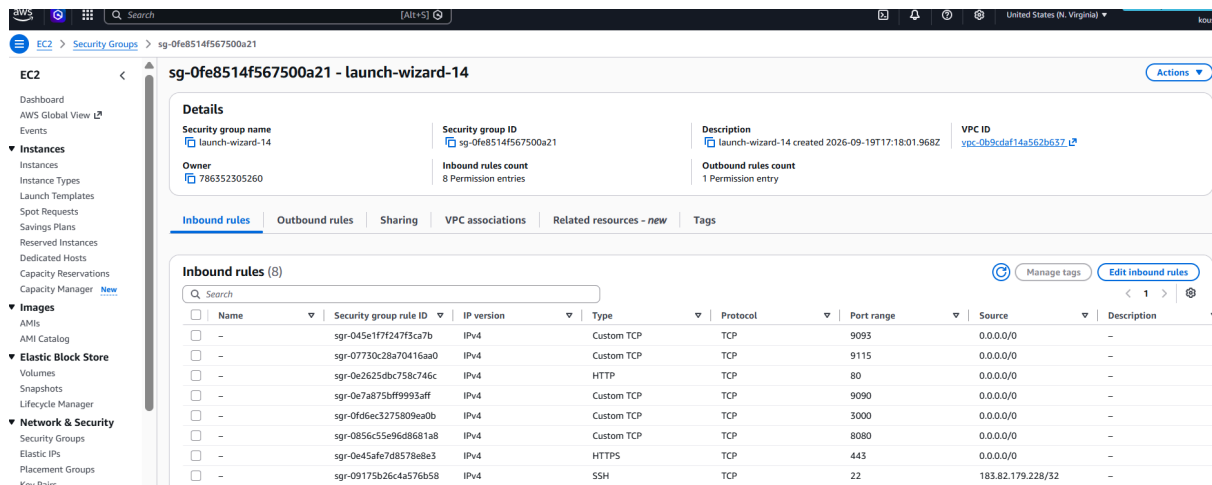
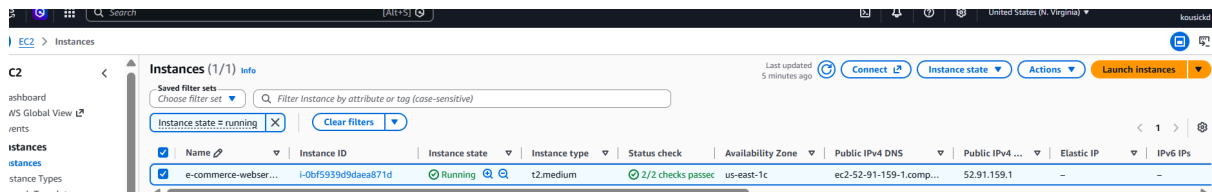
Users((End Users)) -->|HTTP Traffic| App

## Infrastructure Setup

### AWS EC2 & Security Groups

The application and monitoring stack run on an AWS EC2 instance. The associated Security Group requires the following inbound rules:

- **Port 80 (HTTP):** 0.0.0.0/0 (Public application access)
- **Port 22 (SSH):** Restricted to Administrator IP
- **Port 8080 (TCP):** Restricted to Administrator IP (Jenkins UI)
- **Port 9090 (TCP):** Restricted to Administrator IP (Prometheus UI)
- **Port 3000 (TCP):** Restricted to Administrator IP (Grafana UI)



# CI/CD Pipeline Workflow

The deployment relies on a Jenkinsfile executing a Multibranch Pipeline, triggered automatically by GitHub Webhooks on push events.

## Branch Strategies

### 1. dev Branch:

- Code pushed to dev triggers the Jenkins pipeline.
- Jenkins executes `./build.sh` to create the Docker image.
- The image is tagged and pushed to the public dev repository on Docker Hub.
- Jenkins executes `./deploy.sh` to deploy the updated dev container to the server.

```
aws [Q Search] [Alt+S] United States (N. Virginia)
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ git status
On branch dev
Untracked files:
 (use "git add <file>..." to include in what will be committed)
 .dockerignore
 .gitignore
 Dockerfile
 build.sh
 deploy.sh
 docker-compose.yml

nothing added to commit but untracked files present (use "git add" to track)
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ git add .
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ git status
On branch dev
Changes to be committed:
 (use "git restore --staged <file>..." to unstage)
 new file: .dockerignore
 new file: .gitignore
 new file: Dockerfile
 new file: build.sh
 new file: deploy.sh
 new file: docker-compose.yml

ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ git remote -v
origin https://github.com/kousick-devaraj/devops-build.git (fetch)
origin https://github.com/kousick-devaraj/devops-build.git (push)
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ git branch
* dev
main
```

```
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ git commit -m "Add Docker, Compose, and deployment scripts"
[dev 399e6c7] Add Docker, Compose, and deployment scripts
 Committer: Ubuntu <ubuntu@ip-172-31-22-143.ec2.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

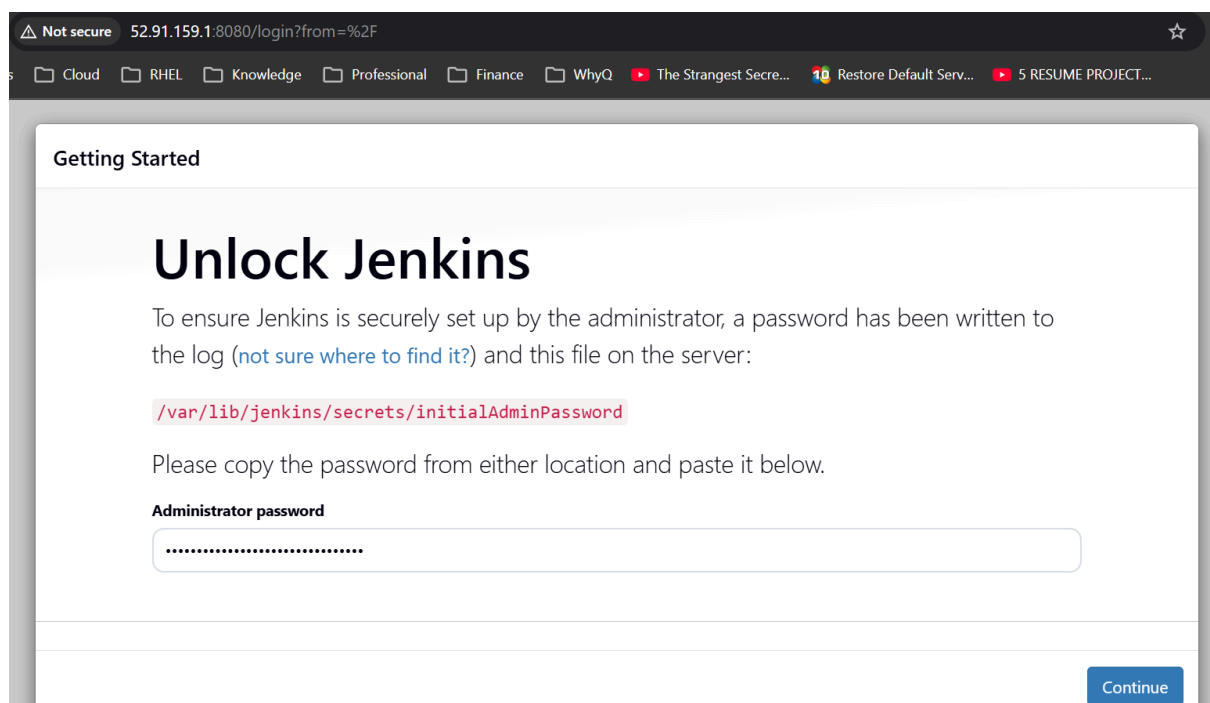
 git config --global --edit

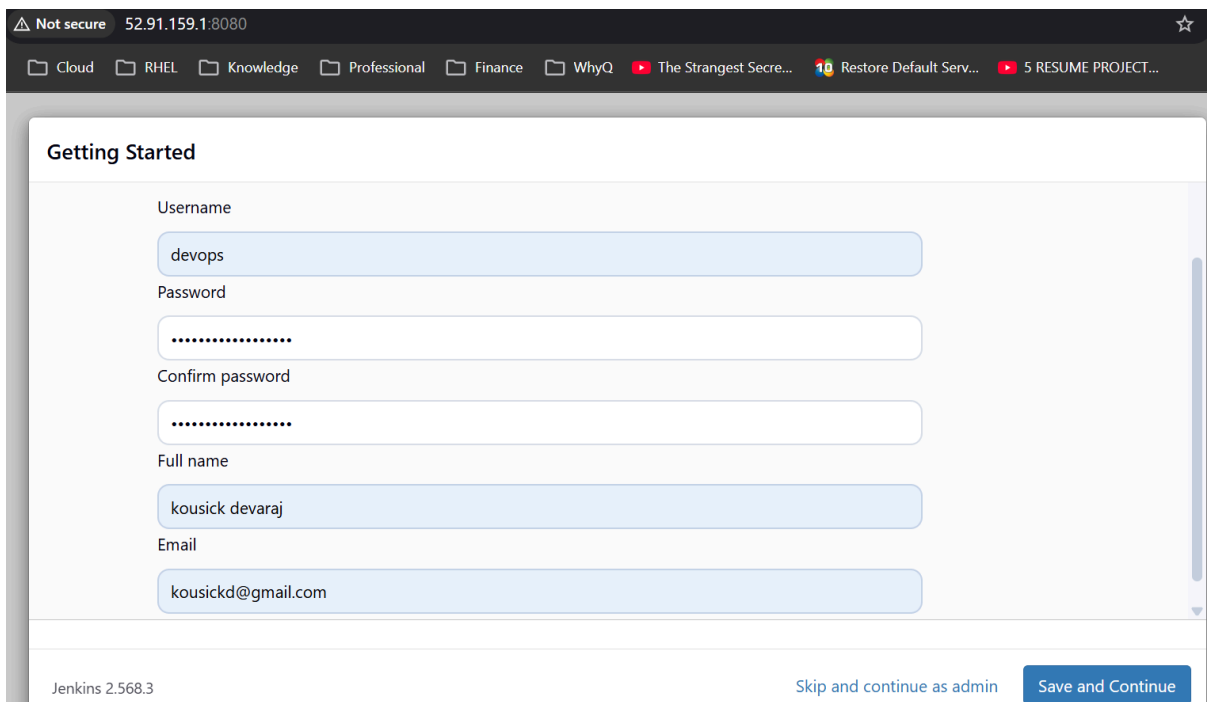
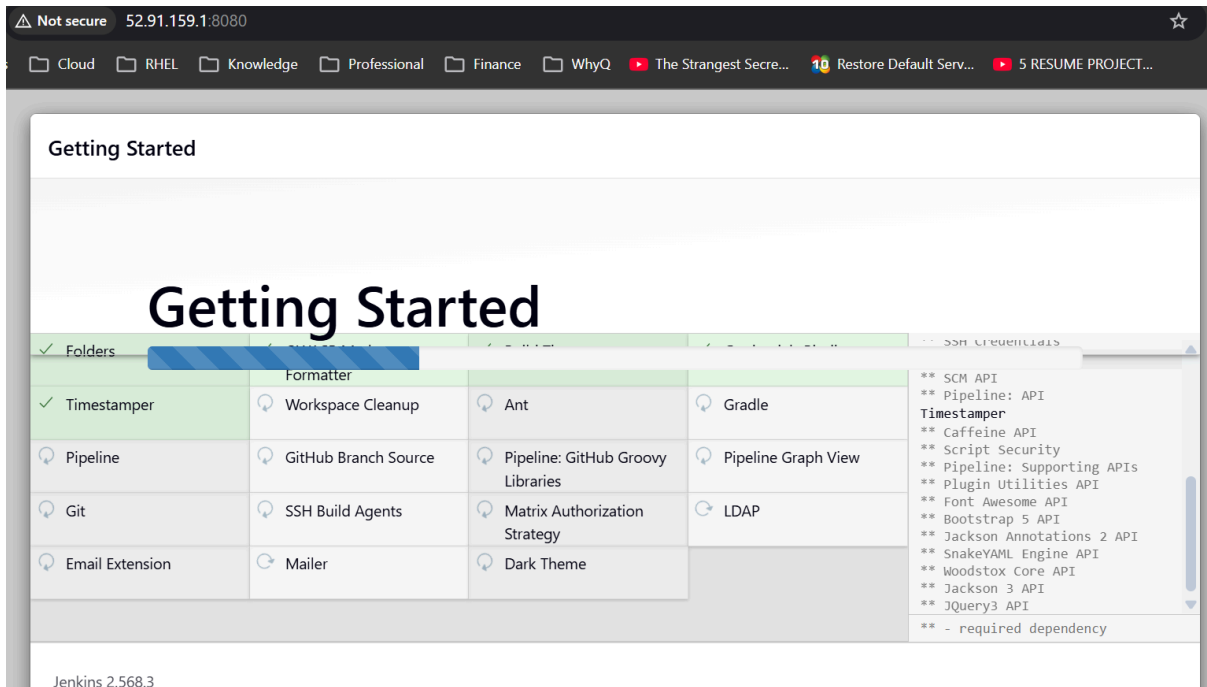
After doing this, you may fix the identity used for this commit with:

 git commit --amend --reset-author

6 files changed, 37 insertions(+)
create mode 100644 .dockerignore
create mode 100644 .gitignore
create mode 100644 Dockerfile
create mode 100644 build.sh
create mode 100644 deploy.sh
create mode 100644 docker-compose.yml
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ git push origin dev
Username for 'https://github.com': kousick-devaraj
Password for 'https://kousick-devaraj@github.com':
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (8/8), 1.04 KiB | 1.04 MiB/s, done.
Total 8 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/kousick-devaraj/devops-build.git
 ef44730..399e6c7 dev -> dev
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$
```

```
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ java -version
openjdk version "21.0.12" 2026-07-21
OpenJDK Runtime Environment (build 21.0.12+8-1-26.04-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.12+8-1-26.04-Ubuntu, mixed mode, sharing)
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ jenkins --version
2.568.3
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ docker-compose --version
docker-compose: command not found
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ docker-compose -version
docker-compose: command not found
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ docker-compose version
docker-compose: command not found
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ docker compose version
Docker Compose version 2.40.3+ds1-0ubuntu1
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$
```





Jenkins URL: <http://3.93.201.150:8080/>

Installed required plugins

DevOpsCloudRHELKnowledgeProfessionalFinanceWhyQThe Strangest Sec...Restore De

Jenkins

Manage Jenkins

Plugins

Plugins

Updates

Available plugins

Installed plugins

Advanced settings

Download progress

Oracle Java SE Development Kit Installer

SSH server

Command Agent Launcher

Docker Compose Build Step

Docker Pipeline

Loading plugin extensions

GitHub Integration

GitHub Authentication

Loading plugin extensions

Success

Success

Success

Success

Success

Success

Success

Success

3.93.201.150:8080/manage/credentials/

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Manage Jenkins

Credentials

Credentials

+ Add Credentials

docker-hub-creds

kousickd/\*\*\*\*\*

System - Global - Creds to connect Docker hub

git-hub-creds

kousick-devaraj/\*\*\*\*\*

System - Global - Creds to connect git repos

Stores scoped to Jenkins

System

Domains: Global

3.93.201.150:8080/job/e-commerce-pipeline/indexing/console

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Jenkins

e-commerce-app

Scan Repository

Configure

Scan Repository Now

Scan Repository Log

View as plain text

Multibranch Pipeline Events

Delete Multibranch Pipeline

Build History

Project Relationship

Check File Fingerprint

Rename

GitHub

Pipeline Syntax

Credentials

Build Queue

No builds in the queue.

Build Executor Status

(0 of 2 executors busy)

Started by user kousick devaraj

[Sun Sep 20 10:49:35 UTC 2026] Starting branch indexing...

10:49:37 Connecting to https://api.github.com using kousick-devaraj/\*\*\*\*\* (Creds to connect git repos)

Examining kousick-devaraj/devops-build

Getting remote pull requests...

Checking branches...

Getting remote branches...

Checking branch main

'Jenkinsfile' not found

Does not meet criteria

Checking branch dev

'Jenkinsfile' found

Met criteria

Changes detected: dev (09d31ae9516c14bef30be8cd17d74939e300394f → 832610c263dbbcaef8748beb8c1c0189030b0345)

Scheduled build for branch: dev

2 branches were processed

Checking pull-requests...

0 pull requests were processed

Finished examining kousick-devaraj/devops-build

[Sun Sep 20 10:49:38 UTC 2026] Finished branch indexing. Indexing took 2.3 sec

```
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ pwd
/home/ubuntu/e-commerce-app/devops-build
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ git add Jenkinsfile
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ git status
On branch dev
Changes to be committed:
 (use "git restore --staged <file>..." to unstage)
 new file: Jenkinsfile

ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ git commit -m "Adding the Jenkinsfile with required configuration"
[dev c97f40c] Adding the Jenkinsfile with required configuration
 Committer: Ubuntu <ubuntu@ip-172-31-22-143.ec2.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

 git config --global --edit

After doing this, you may fix the identity used for this commit with:

 git commit --amend --reset-author

1 file changed, 41 insertions(+)
create mode 100644 Jenkinsfile
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ git push origin dev
Username for 'https://github.com': kousick-devaraj
Password for 'https://kousick-devaraj@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 673 bytes | 673.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/kousick-devaraj/devops-build.git
 339e6d7..c97f40c dev -> dev
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$
```

```
ubuntu@ip-172-31-22-143:~$ sudo docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
7d1e36746e51 kousickd/dev:4 "/docker-entrypoint..." About a minute Up About a minute 0.0.0.0:80->80/tcp, [::]:80->80/tcp react-prod
ubuntu@ip-172-31-22-143:~$
```

Jenkins

e-commerce-app

Status

Configure

Scan Repository Now

Scan Repository Log

Multibranch Pipeline Events

Delete Multibranch Pipeline

Build History

e-commerce-app

Folder name: e-commerce-pipeline

To run react app using containers through CI/CD.

Branches (1) Pull Requests (0)

S	W	Name	Last Success	Last Failure	Last Duration
✓	☁	dev	10 min #4	15 hr #2	11 sec

3.93.201.150:8080/job/e-commerce-pipeline/job/dev/4/console

DevOps Cloud RHEL Knowledge Professional Finance WhyQ The Strangest Sec... Restore Default Serv... 5 RESUME PROJECT...

ins / e-commerce-app / dev / #4

Console

Download Copy View as plain text

Output

Build #4

Overview

Information

From Stage

Steps

Issues

Build

Branch indexing

10:49:47 Connecting to https://api.github.com using kousick-devaraj/\*\*\*\*\* (Creds to connect git repos)

Obtained Jenkinsfile from 832610c263dbbcaef8748beb8c1c0189030b0345

[Pipeline] Start of Pipeline

[Pipeline] node

Running on Jenkins in /var/lib/jenkins/workspace/e-commerce-pipeline\_dev

[Pipeline] {

[Pipeline] stage

[Pipeline] { (Declarative: Checkout SCM)

[Pipeline] checkout

The recommended git tool is: NONE

using credential git-hub-creds

> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/e-commerce-pipeline\_dev/.git # timeout=10

Fetching changes from the remote Git repository

> git config remote.origin.url https://github.com/kousick-devaraj/devops-build.git # timeout=10

Fetching without tags

Fetching upstream changes from https://github.com/kousick-devaraj/devops-build.git

> git --version # timeout=10

> git --version # 'git version 2.53.0'

using GIT\_ASKPASS to set credentials Creds to connect git repos

> git fetch --no-tags --force --progress -- https://github.com/kousick-devaraj/devops-build.git +refs/heads/dev:refs/remotes/origin/dev # timeout=10

Checking out Revision 832610c263dbbcaef8748beb8c1c0189030b0345 (dev)

← → ↺ ⚠ Not secure 3.93.201.150:8080/job/e-commerce-pipeline/job/dev/4/console 🔍 ☆ 📁

📁 DevOps 📁 Cloud 📁 RHEL 📁 Knowledge 📁 Professional 📁 Finance 📁 WhyQ 📁 The Strangest Secre... 📁 Restore Default Serv... 📁 5 RESUME PROJECT...

Jenkins e-commerce-app dev / #4

```
[Pipeline] // script
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Server)
[Pipeline] script
[Pipeline] {
[Pipeline] sh
+ chmod +x deploy.sh
[Pipeline] sh
+ ./deploy.sh kousickd/dev:4
Deploying Docker Image: kousickd/dev:4...
time="2026-09-20T18:49:57Z" level=warning msg="/var/lib/jenkins/workspace/e-commerce-pipeline_dev/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
Container react-prod Stopping
Container react-prod Stopped
Container react-prod Removing
Container react-prod Removed
Network e-commerce-pipeline_dev_default Removing
Network e-commerce-pipeline_dev_default Removed
time="2026-09-20T18:49:58Z" level=warning msg="/var/lib/jenkins/workspace/e-commerce-pipeline_dev/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
Network e-commerce-pipeline_dev_default Creating
Network e-commerce-pipeline_dev_default Created
Container react-prod Creating
Container react-prod Created
Container react-prod Starting
Container react-prod Started
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withenv
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // withenv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline

Github has been notified of this commit's build result

Finished: SUCCESS
```

Docker Hub repository details:

→ ↺ 📁 hub.docker.com/repositories/kousickd 🔍 ☆ 📁

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hub Explore My Hub 🔍 Search Docker Hub CtrlK 📁 🔔

 kousickd  
Your individual account

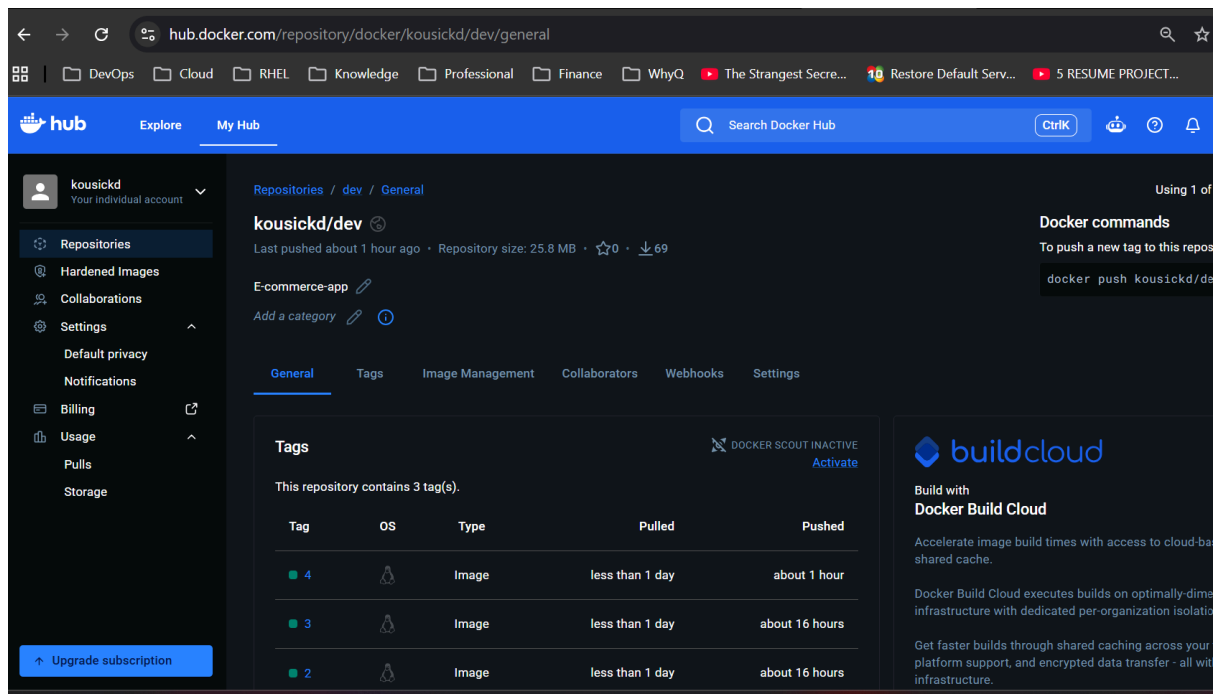
- Repositories
- Hardened Images
- Collaborations
- Settings
  - Default privacy
  - Notifications
- Billing
- Usage
  - Pulls
  - Storage

### Repositories

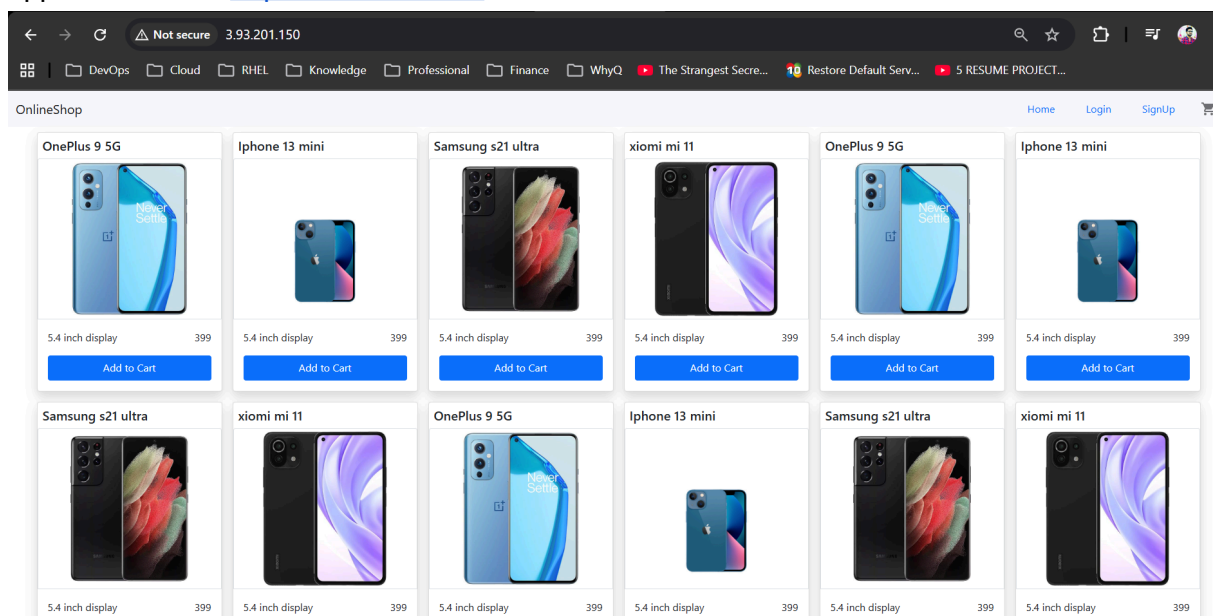
All repositories within the kousickd namespace.

All content

Name	Last Pushed ↑	Contains	Visibility
kousickd/prod	about 18 hours ago	IMAGE	Private
kousickd/dev	about 19 hours ago	IMAGE	Public
kousickd/trend-store-app	7 days ago	IMAGE	Public
kousickd/trendstore	8 days ago	IMAGE	Public
kousickd/demo-app	3 months ago	IMAGE	Public
kousickd/kd-nginx	4 months ago	IMAGE	Public



Application URL: <http://3.93.201.150/>



## 2. master Branch:

- Merging code to master triggers the Jenkins pipeline.
- Jenkins builds and pushes the image to the private prod repository on Docker Hub.
- Jenkins deploys the production-ready container to the server via Docker Compose.

```

ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ ls -lrt
total 24
drwxrwxr-x 3 ubuntu ubuntu 4096 Sep 19 17:35 build
-rw-rw-r-- 1 ubuntu ubuntu 158 Sep 19 17:36 Dockerfile
-rw-rw-r-- 1 ubuntu ubuntu 169 Sep 19 17:37 docker-compose.yml
-rw-rw-r-- 1 ubuntu ubuntu 187 Sep 19 17:38 build.sh
-rw-rw-r-- 1 ubuntu ubuntu 232 Sep 19 19:26 deploy.sh
-rw-rw-r-- 1 ubuntu ubuntu 1360 Sep 20 10:46 Jenkinsfile
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ git branch
 dev
* main
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ git pull origin main
From https://github.com/kousick-devaraj/devops-build
* branch main -> FETCH_HEAD
Already up to date.
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ git merge dev
Updating ef44730..832610c
Fast-forward
 .dockerignore | 4 ++++
 .gitignore | 2 ++
 Dockerfile | 5 +++++
 Jenkinsfile | 41 +++
 build.sh | 8 ++++++++
 deploy.sh | 10 ++++++++
 docker-compose.yml | 8 ++++++++
 7 files changed, 78 insertions(+)
 create mode 100644 .dockerignore
 create mode 100644 .gitignore
 create mode 100644 Dockerfile
 create mode 100644 Jenkinsfile
 create mode 100644 build.sh
 create mode 100644 deploy.sh
 create mode 100644 docker-compose.yml
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$

```

kousick-devaraj / devops-build

Pull requests Actions Projects Wiki Security and quality Insights Settings

**devops-build** Public

forked from [sriram-R-krishnan/devops-build](#)

main 2 Branches 0 Tags

Go to file Add file <> Code

This branch is up to date with [sriram-R-krishnan/devops-build:main](#). Contribute Sync fork

**sriram-R-krishnan** Add files via upload ef44730 · 3 years ago 1 Commit

build Add files via upload 3 years ago

README

us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-0bf59...

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aws Search [Alt+S]

```

ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ git branch
 dev
* main
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$ git push origin main
Username for 'https://github.com': kousick-devaraj
Password for 'https://kousick-devaraj@github.com':
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kousick-devaraj/devops-build.git
 ef44730..832610c main -> main
ubuntu@ip-172-31-22-143:~/e-commerce-app/devops-build$

```

← → ↻ github.com/kousick-devaraj/devops-build/tree/main

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kousick-devaraj / devops-build

<> Code Pull requests Actions Projects Wiki Security and quality Insights Settings

devops-build Public  
forked from [sriram-R-krishnan/devops-build](#)

main 2 Branches 0 Tags

Go to file T Add file <> Code

This branch is 4 commits ahead of [sriram-R-krishnan/devops-build:main](#) . Contribute Sync fork

Ubuntu Update repo name from master to main 832610c · 47 minutes ago 5 Commits

build	Add files via upload	3 years ago
.dockerignore	Add Docker, Compose, and deployment scripts	17 hours ago
.gitignore	Add Docker, Compose, and deployment scripts	17 hours ago
Dockerfile	Add Docker, Compose, and deployment scripts	17 hours ago
Jenkinsfile	Update repo name from master to main	47 minutes ago
build.sh	Add Docker, Compose, and deployment scripts	17 hours ago
deploy.sh	updated docker-compose command in deploy.sh	16 hours ago
docker-compose.yml	Add Docker, Compose, and deployment scripts	17 hours ago

← → ↻ ⚠ Not secure 3.93.201.150:8080/job/e-commerce-pipeline/job/main/1/console

DevOps Cloud RHEL Knowledge Professional Finance WhyQ The Strangest Secre... 10 Restore Default Serv... 5 RESUME PROJECT...

Jenkins / e-commerce-app / main / #1

Status Console Download

Changes

Console Output

Delete build '#1'

Git Build Data

Pipeline Overview

Edit build information

Restart from Stage

Replay

Pipeline Steps

Workspaces

```
Push event to branch main
11:33:47 Connecting to https://api.github.com using kousick-devaraj/***** (Creds to connect git repos)
Obtained Jenkinsfile from 832610c263dbbcaef8748beb8c1c0189030b0345
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/e-commerce-pipeline_main
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
The recommended git tool is: NONE
using credential git-hub-creds
Cloning the remote Git repository
Cloning with configured refspecs honoured and without tags
Cloning repository https://github.com/kousick-devaraj/devops-build.git
> git init /var/lib/jenkins/workspace/e-commerce-pipeline_main # timeout=10
Fetching upstream changes from https://github.com/kousick-devaraj/devops-build.git
> git --version # timeout=10
> git --version # 'git version 2.53.0'
using GIT_ASKPASS to set credentials Creds to connect git repos
> git fetch --no-tags --force --progress -- https://github.com/kousick-devaraj/devops-build.git +refs/heads/main:refs/remotes/origin/main # timeout=10
> git config remote.origin.url https://github.com/kousick-devaraj/devops-build.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/main:refs/remotes/origin/main # timeout=10
Avoid second fetch
Checking out Revision 832610c263dbbcaef8748beb8c1c0189030b0345 (main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 832610c263dbbcaef8748beb8c1c0189030b0345 # timeout=10
Commit message: "Update repo name from master to main"
First time build. Skipping changelog.
```

```
3.93.201.150:8080/job/e-commerce-pipeline/job/main/1/console

+ chmod +x build.sh
[Pipeline] sh
+ ./build.sh kousickd/prod:1
Building Docker image: kousickd/prod:1...
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 2.623MB

Step 1/4 : FROM nginx:alpine
--> 62ff2089abf5
Step 2/4 : COPY build /usr/share/nginx/html
--> Using cache
--> 7c48dc5b5745
Step 3/4 : EXPOSE 80
--> Using cache
--> 32747622c45a
Step 4/4 : CMD ["nginx", "-g", "daemon off;"]
--> Using cache
--> 5371bd1b1504
Successfully built 5371bd1b1504
Successfully tagged kousickd/prod:1
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Push Image)
[Pipeline] script
[Pipeline] {
[Pipeline] sh
+ echo ****
```

hub.docker.com/repository/docker/kousickd/prod/general

hub Explore My Hub Search Docker Hub CtrlK

kousickd Your individual account

Repositories Hardened Images Collaborations Settings Default privacy Notifications Billing Usage Pulls Storage

Repositories / prod / General

kousickd/prod Last pushed 7 minutes ago · Repository size: 25.8 MB · ☆0 · ↓11

E-commerce-app Add a category

General Tags Image Management Collaborators Webhooks Settings

Tags DOCKER SCOUT INACTIVE Activate

This repository contains 1 tag(s).

Tag	OS	Type	Pulled	Pushed
1		Image	less than 1 day	7 minutes

Build with Docker Build Cloud Accelerate image build times with shared cache. Docker Build Cloud executes builds on infrastructure with dedicated resources.

### Monitoring:

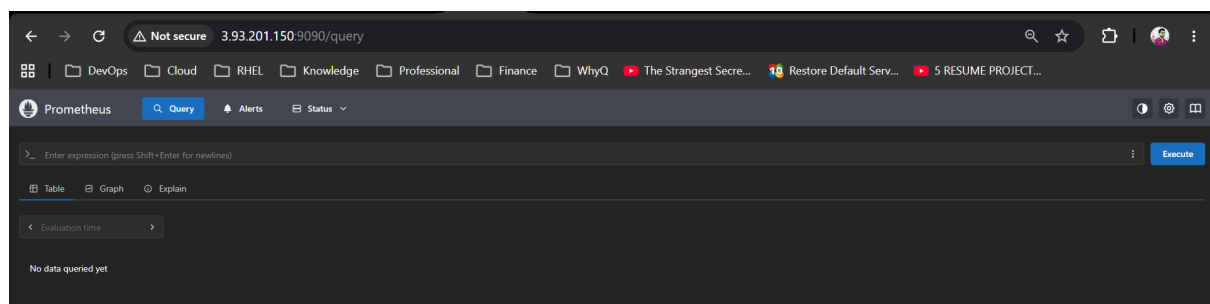
1. Setup a monitoring system to check the health status of the application. (Prometheus and grafana)
2. Sending notifications only if the application goes down is highly appreciable

```

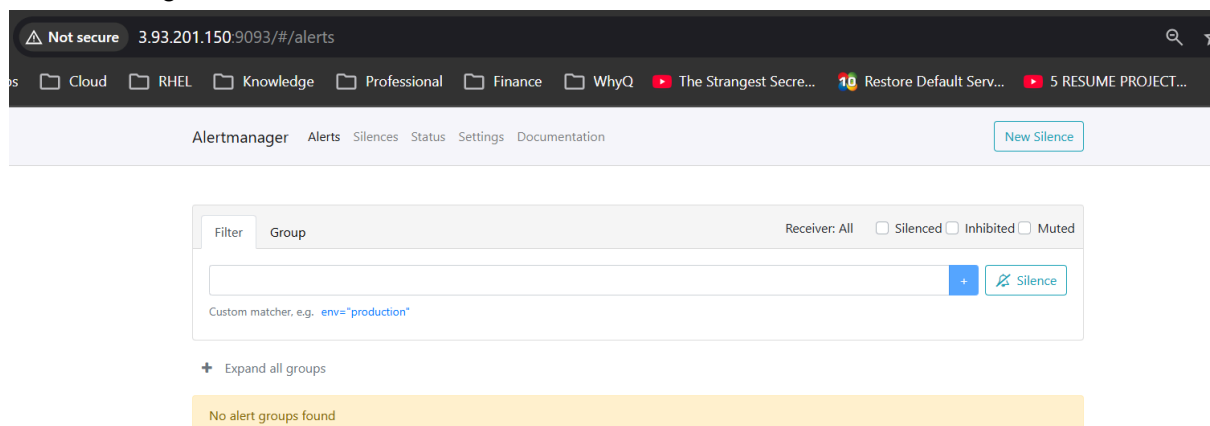
description -nms docker app at [[ipaddress.hostname]] has been successfully installed.
ubuntu@ip-172-31-22-143:/e-commerce-app/devops-build/monitoring$ sudo docker-compose up -d
sudo docker-compose: command not found
ubuntu@ip-172-31-22-143:/e-commerce-app/devops-build/monitoring$ sudo docker compose up -d
[WARN] [0000] /home/ubuntu/e-commerce-app/devops-build/monitoring/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
#
Container monitoring-alertmanager-1 Running 0.0s
Container monitoring-blackbox-exporter-1 Running 0.0s
Container monitoring-grafana-1 Running 0.0s
Container monitoring-prometheus-1 Stopped 0.0s
ubuntu@ip-172-31-22-143:/e-commerce-app/devops-build/monitoring$ docker ps
permission denied while trying to connect to the Docker API at unix:///var/run/docker.sock
ubuntu@ip-172-31-22-143:/e-commerce-app/devops-build/monitoring$ sudo docker compose ps
[WARN] [0000] /home/ubuntu/e-commerce-app/devops-build/monitoring/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
NAME IMAGE COMMAND SERVICE CREATED STATUS PORTS
monitoring-alertmanager-1 prom/alertmanager "/bin/alertmanager ..." alertmanager 26 minutes ago Up 10 minutes 0.0.0.0:9093->9093/tcp, :::9093->9093/tcp
monitoring-blackbox-exporter-1 prom/blackbox-exporter "/bin/blackbox_export..." blackbox-exporter 26 minutes ago Up 26 minutes 0.0.0.0:9115->9115/tcp, :::9115->9115/tcp
monitoring-grafana-1 grafana/grafana "run.sh" grafana 26 minutes ago Up 26 minutes 0.0.0.0:3000->3000/tcp, :::3000->3000/tcp
monitoring-prometheus-1 prom/prometheus "/bin/prometheus --c..." prometheus 26 minutes ago Up 34 seconds 0.0.0.0:9090->9090/tcp, :::9090->9090/tcp

```

## Prometheus:



Alerts manager:



# Monitoring validation:

## Docker service stopped and alert triggered

```
*** System restart required ***
Last login: Sun Sep 20 12:18:14 2026 from 18.206.107.27
ubuntu@ip-172-31-22-143:~$ sudo docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
44ad6dc7436 prom/prometheus "/bin/prometheus --c..." 49 minutes ago Up 24 minutes 0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp monitoring-prometheus-1
7a5f6eb1a8f0 grafana/grafana "/run.sh" 49 minutes ago Up 45 minutes 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp monitoring-grafana-1
3b1e55982a9 prom/alertmanager "/bin/alertmanager --..." 49 minutes ago Up 34 minutes 0.0.0.0:9093->9093/tcp, [::]:9093->9093/tcp monitoring-alertmanager-1
865c4d74e1e prom/blackbox-exporter "/bin/blackbox_expore..." 49 minutes ago Up 49 minutes 0.0.0.0:9115->9115/tcp, [::]:9115->9115/tcp monitoring-blackbox-exporter-1
7d1e36746e51 kousickd/dev:4 "/docker-entrypoint..." 2 hours ago Up 2 hours 0.0.0.0:80->80/tcp, [::]:80->80/tcp react-prod

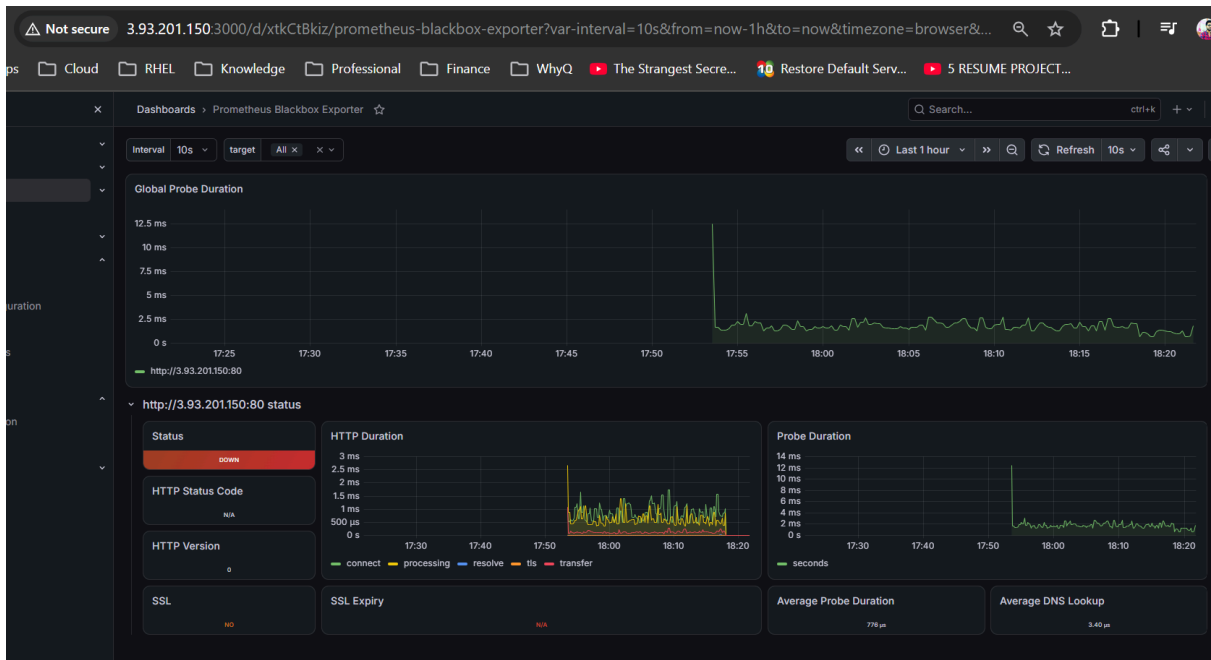
ubuntu@ip-172-31-22-143:~$ sudo docker stop react-prod
permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
ubuntu@ip-172-31-22-143:~$ sudo docker stop react-prod
react-prod
ubuntu@ip-172-31-22-143:~$ sudo docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
44ad6dc7436 prom/prometheus "/bin/prometheus --c..." 50 minutes ago Up 23 minutes 0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp monitoring-prometheus-1
7a5f6eb1a8f0 grafana/grafana "/run.sh" 50 minutes ago Up 50 minutes 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp monitoring-grafana-1
3b1e55982a9 prom/alertmanager "/bin/alertmanager --..." 50 minutes ago Up 34 minutes 0.0.0.0:9093->9093/tcp, [::]:9093->9093/tcp monitoring-alertmanager-1
865c4d74e1e prom/blackbox-exporter "/bin/blackbox_expore..." 50 minutes ago Up 50 minutes 0.0.0.0:9115->9115/tcp, [::]:9115->9115/tcp monitoring-blackbox-exporter-1
7d1e36746e51 kousickd/dev:4 "/docker-entrypoint..." 2 hours ago Exited (0) 13 seconds ago react-prod
ubuntu@ip-172-31-22-143:~$ []
```

The screenshot shows the Prometheus Alerts page in a web browser. The URL is `3.93.201.150:9090/alerts`. The page displays a single alert in a 'PENDING (1)' state. The alert is titled 'ApplicationDown' and is associated with the rule `/etc/prometheus/alerts.yml`. The alert details include:

- ApplicationDown** (PENDING (1))
- probe\_success == 0**
- for: 1m**
- severity: critical**
- description:** The React app at [ [labels.instance]] has been unreachable for more than 1 minute.
- summary:** React Application is DOWN
- Alert labels:** alertname="ApplicationDown", instance="http://3.93.201.150:80", job="react\_app\_status", severity="critical"
- State:** PENDING
- Active Since:** 40.812s
- Value:** 0

The screenshot shows a Gmail inbox with an email notification for a firing alert. The email is from `kousickd@gmail.com` and is titled `[FIRING:1] (ApplicationDown http://3.93.201.150:80 react_app_status critical)`. The email content includes:

- 1 alert for**
- View In Alertmanager** button
- [1] Firing**
- Labels:** alertname = ApplicationDown, instance = <http://3.93.201.150:80>, job = react\_app\_status, severity = critical
- Annotations:** description = The React app at <http://3.93.201.150:80> has been unreachable for more than 1 minute. summary = React Application is DOWN



Docker service started and alert resolved:

```
Last login: Sun Sep 20 12:47:15 2026 from 18.206.107.29
ubuntu@ip-172-31-22-143:~$ sudo docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
44ad6edc7436 prom/prometheus "/bin/prometheus --c..." 58 minutes ago Up 43 minutes 0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp monitoring-prometheus-1
7a5f8eb1a8f0 grafana/grafana "/run.sh" 58 minutes ago Up 58 minutes 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp monitoring-grafana-1
3b1e855982e9 prom/alertmanager "/bin/alertmanager -..." 58 minutes ago Up 43 minutes 0.0.0.0:9093->9093/tcp, [::]:9093->9093/tcp monitoring-alertmanager-1
865ca4d74e1e prom/blackbox-exporter "/bin/blackbox_export..." 58 minutes ago Up 58 minutes 0.0.0.0:9115->9115/tcp, [::]:9115->9115/tcp monitoring-blackbox-exporter-1
741e36746e51 koushick/dev4 "/docker-entrypoint..." 2 hours ago Exited (0) 8 minutes ago react-prod

ubuntu@ip-172-31-22-143:~$ sudo docker start react-prod
react-prod
ubuntu@ip-172-31-22-143:~$
ubuntu@ip-172-31-22-143:~$ sudo docker start react-prod
react-prod
ubuntu@ip-172-31-22-143:~$ sudo docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
44ad6edc7436 prom/prometheus "/bin/prometheus --c..." 59 minutes ago Up 43 minutes 0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp monitoring-prometheus-1
7a5f8eb1a8f0 grafana/grafana "/run.sh" 59 minutes ago Up 59 minutes 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp monitoring-grafana-1
3b1e855982e9 prom/alertmanager "/bin/alertmanager -..." 59 minutes ago Up 43 minutes 0.0.0.0:9093->9093/tcp, [::]:9093->9093/tcp monitoring-alertmanager-1
865ca4d74e1e prom/blackbox-exporter "/bin/blackbox_export..." 59 minutes ago Up 59 minutes 0.0.0.0:9115->9115/tcp, [::]:9115->9115/tcp monitoring-blackbox-exporter-1
741e36746e51 koushick/dev4 "/docker-entrypoint..." 2 hours ago Up 28 seconds 0.0.0.0:80->80/tcp, [::]:80->80/tcp react-prod
ubuntu@ip-172-31-22-143:~$
```

